KHANDELWAL COLLEGE OF MANAGEMENT AND TECHNOLOGY, BAREILLY



SESSION: 2024-2025

SYNOPSIS

OF

PROJECT

ON

Zesty Hub

SUBMITTED TO:

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📄 Zesty Hub Project Documentation

1. INTRODUCTION

Zesty Hub is an interactive web-based collaboration platform designed to enhance real-time communication among users. It offers features like secure user authentication, private and group chats, video calling, and a collaborative online IDE for document and code writing. Built using modern technologies like React.js, Node.js, Firebase, and WebRTC, it provides a seamless experience for developers, students, and professional teams.

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2. OBJECTIVE

The main objective of Zesty Hub is to create an all-in-one platform where users can:

- Authenticate securely using Email/Password or Google.
- Connect with friends, chat in real-time, and share resources.
- Make high-quality peer-to-peer video calls.
- Collaborate on projects by writing and sharing code and documents.
- Create or join groups for larger discussions and collaborations.

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3. SOFTWARE AND HARDWARE TOOLS

Software Tools

- **Frontend**: React.js, Tailwind CSS
- **Backend**: Node.js, Express.js
- **Database**: Firebase Firestore
- **Authentication**: Firebase Authentication
- **Video Calling**: WebRTC (custom signaling server)

- **Version Control**: GitHub

Hardware Requirements

- Any standard laptop/desktop with internet access.
- Minimum 8 GB RAM and a modern web browser (Google Chrome, Firefox, Edge).
- Webcam and microphone for video calling features.

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4. MODULES OF PROJECT

- **Authentication Module**: Handles user sign-up, login, and Google sign-in.
- **Dashboard Module**: Displays user profile, friends list, groups, and settings.
- **Chat Module**: Enables one-on-one and group chat functionalities.
- **Video Calling Module**: Allows real-time video and audio calls using WebRTC.
- **Collaborative IDE Module**: Enables users to create, edit, and collaborate on documents and code.
- **Group Management Module**: Lets users create new groups, invite/join members, and manage group activities.

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5. ROLES & RESPONSIBILITIES OF TEAM MEMBERS

Team Member	Role	Responsibility
Divyansh Chauhan	Frontend Developer	Build UI components using React and Tailwind, implement user interaction features
Amir Hussain	Backend Developer	Develop and maintain the Node.js and Express.js server, manage API endpoints
Priyanshu Bilhatiya	Database Manager	Design and maintain the Firestore database, implement Firebase security rules

Sameer Javed	WebRTC Engineer	Setup and manage WebRTC signaling server for real-time communication

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6. FUTURE ENHANCEMENT

- Adding screen sharing and recording functionality during video calls.
- Integration with GitHub for project collaboration.
- Mobile application version using React Native.
- Enhanced security features such as Two-Factor Authentication (2FA).
- Implementing AI-driven chatbots for group support.
- Push notifications for incoming chats and calls.

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7. CONCLUSION

Zesty Hub brings together the essential elements of real-time collaboration on a single platform. It offers users a streamlined experience for chatting, video calling, and project collaboration, empowering teams and individuals to stay connected and productive. The use of modern, scalable technologies ensures the system can grow and adapt to future user needs.

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8. REFERENCE

- React.js Official Documentation
- Node.js Official Documentation
- Firebase Documentation
- WebRTC Documentation
- Tailwind CSS Documentation

- Express.js Documentation